

LOTTE MART VN - ONLINE RENEWAL 2026

Test Plan — Part 1

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APPROVAL

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CHANGE HISTORY

Version	Date	Author	Section	Description
v0.1	2026-08-13	Hien Nguyen	All	Master Test Plan created from the Part 1 Feature Baseline. Timeline, test environment and test tooling left as TBD.
v0.2	2026-08-14	Hien Nguyen	All	Re-issued in the MART v3.0 document format (English only). RECIPE — AI Recipe Shopping removed from Part 1 scope; Part 1 is now 6 modules / 35 feature groups / 64 features. Open Items section added for every TBD.
v0.3	2026-08-17	Hien Nguyen	2, 11, 13	Test scope extended with two screen-level sub-sections: 2.8 Smart Search (from MART-DE03-Smart Search Storyboard-Mobile App-v0.1-20260816) and 2.9 Simplified User Flow (from MART-AN02-SRS-Simplify User Flow-v0.1-20260814). Traceability 11.2 / 11.3 and Open Items extended accordingly. Document format unchanged — template v3.0.

PURPOSE

This document is the consolidated Test Plan for the Part 1 scope of the MART VN Online Renewal 2026 programme. It defines the test objectives, scope, approach, environment, entry and exit criteria, schedule, risks and requirement traceability for the six Part 1 modules, and is the basis for authoring and executing the test cases recorded in TE02 Integration Test Result.

RELATED DOCUMENTS

- AN01 Requirements Definition — requirement baseline per module.

- MART Online Renewal 2026 Scope and Tasks — Part 1 feature baseline, source of the requirement list below.
- DE03 Screen List & Design and DE04 UI Storyboard — screens referenced by the functional test cases.
- MART-AN02-SRS-Simplify User Flow-v0.1-20260814 — screen-level requirement specification for the Simplified User Flow module; source of section 2.9.
- MART-DE03-Smart Search Storyboard-Mobile App-v0.1-20260816 — screen-level storyboard for the Smart Search module; source of section 2.8.
- TE02 Integration Test Result — execution results and defect status produced against this plan.
- PM05 Issue & Risk Register — where test risks are escalated once they materialise.

TABLE OF CONTENTS

1. Introduction & Test Objectives	3
2. Scope.....	4
2.1 Scope Summary	4
2.2 AISCH — AI Search	4
2.3 AIREC — AI Recommendation	5
2.4 SUBS — Subscription Service	5
2.5 HOME — Personalized Homepage	5
2.6 SUF — Simplified User Flow.....	6
2.7 SOCIAL — Social Commerce	6
2.8 Smart Search — Screen-Level Test Scope (AISCH detail).....	6
2.8.1 Search modes in scope	6
2.8.2 Screen-level test scope	7
2.8.3 Rule-based test conditions	8
2.8.4 Out of scope for Smart Search.....	8
2.9 Simplified User Flow — Requirement-Level Test Scope (SUF detail).....	8
2.9.1 Business areas in scope	9
2.9.2 Process-flow test scenarios.....	9
2.9.3 Branch and negative test conditions	9
2.9.4 Business-rule assertion groups	10
2.9.5 Screen inventory.....	10
2.9.6 Non-functional requirements in scope.....	11
2.9.7 Out of scope for Simplified User Flow	11
2.10 Out of Scope.....	11
3. Assumptions	11
4. Dependencies.....	12
5. Test Environment	12
6. Test Approach	12
7. Entry & Exit Criteria	13
7.1 Entry Criteria.....	13
7.2 Exit Criteria	13
8. Test Schedule.....	13
9. Risks & Mitigation	14
10. Test Deliverables.....	14
11. Requirements & Traceability	14
11.1 Part 1 modules — feature-level requirements	14
11.2 Smart Search — screen-level coverage (source: DE03 Storyboard v0.1)	16
11.3 Simplified User Flow — requirement-level coverage (source: AN02 SRS v0.1)	16

12. Roles & Responsibilities 17

13. Open Items 17

1. Introduction & Test Objectives

This is the consolidated Test Plan for the six modules that make up Part 1 of the Feature Baseline: AISCH (AI Search), AIREC (AI Recommendation), SUBS (Subscription Service), HOME (Personalized Homepage), SUF (Simplified User Flow) and SOCIAL (Social Commerce). It summarises scope, approach, schedule, risk and requirement traceability at module and feature-group level.

Two of the six modules now have a screen-level specification. Smart Search is specified in MART-DE03-Smart Search Storyboard-Mobile App-v0.1-20260816 and the Simplified User Flow in MART-AN02-SRS-Simplify User Flow-v0.1-20260814. For those two modules the test scope in this plan is stated at screen and requirement level (sections 2.8 and 2.9); the remaining four modules stay at feature level until their own specifications are issued.

The objectives of this test effort are to:

- Verify that each in-scope requirement, across all six modules, is correctly implemented and traceable to one or more executed test cases.
- Confirm functional behaviour, input validation and business-rule enforcement under positive and negative conditions for every module.
- Exercise boundary, exception and error-handling paths for every module.
- Validate AI-generated and personalized output — relevance, ranking and recommendation accuracy — where a module includes such features.
- Establish an execution-ready, auditable evidence trail for stakeholder sign-off across the whole Part 1 scope.

2. Scope

2.1 Scope Summary

Code	Module	Feature groups	Features	Platform
AISCH	AI Search	6	12	Web / mWeb / App
AIREC	AI Recommendation	6	11	Web / mWeb / App
SUBS	Subscription Service	8	15	Web / mWeb / App + Admin
HOME	Personalized Homepage	10	12	Web / mWeb / App + Admin
SUF	Simplified User Flow	2	7	Web / mWeb / App
SOCIAL	Social Commerce	3	7	App + Admin
Total	6 modules	35	64	—

2.2 AISCH — AI Search

12 features · 6 feature groups

Feature group	Feature	Priority
Search Intent	Natural language query understanding	High
Visual Search	Take an Image — camera-based product search	High
Visual Search	Upload an Image — image upload-based product search	High
Visual Search	Search by Image — AI-powered visual product matching	High
Visual Search	Record Voice — voice input for product search	High
Visual Search	Search by Voice — speech-to-text product search	High
Multi-language Search	Cross-language product search	Medium
Search Result	Display Results — structured search result display	High
Search Result	Default Ranking — category-based ranking algorithm	High
Search Result Sponsor Display	Paid search placement with keyword targeting	High
Product Recommendation	View Suggestions — personalized product suggestions	Medium

Product Recommendation	AI Recommend — AI-driven personalized product recommendation	Medium
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2.3 AIREC — AI Recommendation

11 features · 6 feature groups

Feature group	Feature	Priority
AI Product Review Summary	View Summary — at-a-glance review summary on the PDP	High
AI Product Review Summary	Summarize — AI-powered review aggregation and summarization	High
Notification Suggestion	Send Notification — behaviour-triggered personalized push notifications	Medium
Cart Cross-sell	View Complementary — cart-page complementary product display	Medium
Cart Cross-sell	AI Recommend Complementary — AI-powered basket-based cross-sell	Medium
Coupons Suggestion	View Coupons — contextual coupon display on the cart	Medium
Coupons Suggestion	AI Recommend Coupons — personalized coupon recommendation	Medium
Product Suggestion	Input Requests — free-text shopping request input	High
Product Suggestion	AI Process & Recommend — intent-aware AI product recommendation	High
Shopping Suggestion	Select Shopping Idea — lifestyle-based shopping idea suggestions	High
Shopping Suggestion	AI Recommend Shopping Ideas — AI-generated personalized shopping inspiration	High

2.4 SUBS — Subscription Service

15 features · 8 feature groups

Feature group	Feature	Priority
Subscription Signup & Payment	Introduce Service — subscription landing and onboarding page	High
Subscription Signup & Payment	Select Plan — plan selection with terms and conditions agreement	High
Subscription Signup & Payment	Execute Payment — multi-method subscription payment processing	High
Subscription Activation	Display Dashboard — subscriber My Page with plan and benefit overview	High
Subscription Activation	Apply Subscription Benefits — automatic benefit application at checkout	High
Subscription Management	Manage Payment Method — self-serve payment method management	High
Subscription Management	Change Plan — plan upgrade and downgrade capability	High
Subscription Management	Cancel Subscription — cancellation with retention flow	High
Subscription Auto-Renewal	Recurring Payment — automated recurring subscription billing	High
Subscription Auto-Renewal	Payment Failure Handling — retry logic with escalation	High
Subscription Benefits	Manage Plan — admin subscription plan configuration	High
Subscription Benefits	Configure Bonus Points — subscriber-exclusive bonus point configuration	High
Subscriber List Management	View Subscriber List — admin subscriber list view	High
Subscription Refund Management	Auto-Refund — automated subscription refund processing	High
Subscription Notification	Trigger In-web / App — event-triggered in-app subscription notifications	High

2.5 HOME — Personalized Homepage

12 features · 10 feature groups

Feature group	Feature	Priority
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Hero Welcome + Search	Welcome Greeting — personalized homepage greeting by user segment	High
Hero Welcome + Search	Search Entry — prominent search bar with quick keyword suggestions	High
Personalized Recommendation Feed	View Products — personalized product feed on the homepage	High
Personalized Recommendation Feed	Buy Again — repeat purchase shortcut module	High
Deals / Price Shock Section	High-visibility deals with urgency mechanics	High
Membership CTA Block	Guest-targeted membership acquisition block	High
Content (Live / Video)	Shoppable Live / Video — shoppable live and video content module	High
Meal / Lifestyle Inspiration	Recipe Bundle — recipe-to-cart inspiration bundle on the homepage	High
Personalized Banner Recommendation	Segment-based personalized banner display	High
Personalized Product Recommendation	AI-powered homepage product recommendation	Medium
Homepage Structure & Zone Management	Manage Zones — flexible homepage zone management	Medium
Content Scheduling & Priority	Time-based Scheduling — campaign-based content scheduling	High

2.6 SUF — Simplified User Flow

7 features · 2 feature groups

Feature group	Feature	Priority
Simplified Purchase Process	Single-page Checkout — one-page streamlined checkout flow	High
Simplified Purchase Process	Delivery Speed Selector — one-tap delivery speed selection	High
Simplified Purchase Process	Essential Add-ons — context-aware add-on product carousel at checkout	High
Simplified Purchase Process	Sticky Place Order CTA — always-visible sticky checkout CTA	High
Simplified Signup	Quick Signup — phone-only quick signup	High
Simplified Signup	Zalo Availability Check — smart OTP channel selection (Zalo / SMS)	High
Simplified Signup	OTP Verification — phone-based OTP identity verification	High

2.7 SOCIAL — Social Commerce

7 features · 3 feature groups

Feature group	Feature	Priority
Short-Video Feed	Display Video List — vertical short-video feed	High
Short-Video Feed	Display Products Tagged — shoppable product tags on video	High
Short-Video Feed	Add-to-cart Product Tagged — in-video add-to-cart	High
Short-Video Feed	Share Video — social sharing for viral product discovery	Should
Short-Video Management	Upload Video — admin / seller video content upload	High
Short-Video Management	Attach Product to Video — product tagging management for videos	High
Short-Video Analytics	View Tracking Reports — video performance analytics dashboard	High

2.8 Smart Search — Screen-Level Test Scope (AISCH detail)

Section 2.2 states the AISCH scope at feature level, taken from the Feature Baseline. Since that baseline was written, the module has been specified screen by screen in MART-DE03-Smart Search Storyboard-Mobile App-v0.1-20260816. This section restates the AISCH test scope at screen level so test cases can be authored directly against the storyboard. Where section 2.2 and this section differ, this section governs.

2.8.1 Search modes in scope

Mode	Screen IDs	Screens	What is tested
Text search	SS-SCR-001 ... 012	12	Pre-search, autocomplete, spell correction, results (grid / list), cross-sell, sort, filter, post-filter, add-to-cart, zero result
Image search	SS-SCR-013 ... 016	4	Camera capture, gallery selection, recognition results, unrecognised-image error
Voice search	SS-SCR-017 ... 019	3	Listening state, voice-to-result, unclear-speech error
Shared components	SS-SCR-004 · SS-SCR-011	(2)	Processing / loading state and add-to-cart toast are reused by all three modes — regression must cover all three entry paths after any change

19 distinct screens and states. Platform under test: Mobile App (iOS / Android), 390 × 852 reference frame. The Web variant inherits the same logic with a different layout and is not covered by the storyboard — see 2.8.4.

2.8.2 Screen-level test scope

Screen ID	Screen	REQ ID	Test focus	Test types
SS-SCR-001	Search — Pre-search state	FR-SP-02	Omni-search bar, recent-keyword chips (max 10), trending keywords (Top 10), "Frequently purchased" block, quick-add (+)	Functional · Validation · Negative
SS-SCR-002	Search — Autocomplete	FR-SP-01	Suggestion list (max 10), match highlighting, 300 ms debounce, empty-response behaviour, tap vs Enter	Functional · Validation · Performance
SS-SCR-003	Search — Spell correction	FR-SP-03	Fuzzy-match banner, corrected-keyword link, no-auto-replace rule, suggestions on normalised keyword	Functional · Negative
SS-SCR-004	Search — Processing (shared)	NFR-SS-01	Shown when response > 300 ms; network-error state with Retry when > 5 s; keyword / recognition chip echoed back; never a blank screen	Exception · Performance
SS-SCR-005	Search results — Grid view	FR-PR-01 / FR-PR-02	Relevance ranking, result count, product card contents, Sponsored label, view toggle, search bar retains keyword and Mic / Camera	Functional · Validation
SS-SCR-006	Search results — List view	FR-PR-02	Same result set and ordering as grid; layout switch preserves scroll position, filters and sort	Functional · Integration
SS-SCR-007	Search results — "You may also need" cross-sell	FR-PR-05	Cross-sell block placement, source of the suggested set, add-to-cart from the block	Functional
SS-SCR-008	Search results — Sort	FR-SF-01	Sort options, single selection, result reordering, interaction with active filters	Functional · Validation
SS-SCR-009	Search results — Filter sheet	FR-SF-02	Filter attributes, multi-select, Apply / Reset, count of active filters	Functional · Validation · Negative
SS-SCR-010	Search results — After filter applied	FR-SF-03	Filter chips, result count update, removing one filter, filter persistence on a new keyword (open item)	Functional · Negative
SS-SCR-011	Add to cart from results (shared)	FR-AC-01	Quantity change, cart badge update, confirmation toast and its action button, add from grid / list / cross-sell / image / voice results	Functional · Integration
SS-SCR-012	Zero result	FR-ZR-02	Three-tier recovery: spell correction → substitute and cross-sell products → best sellers. No blank page in any path	Functional · Negative · Exception
SS-SCR-013	Image search — Camera capture	FR-IS-01	Camera permission, capture, retake, switch to gallery	Functional · Negative
SS-SCR-	Image search —	FR-IS-02	Photo-library permission, selection,	Functional ·

014	Gallery selection		unsupported format and oversize handling	Validation · Negative
SS-SCR-015	Image search — Recognition results	FR-IS-03	Recognised item echoed back, result set, same sort / filter / add-to-cart behaviour as text results	Functional · Integration
SS-SCR-016	Image search — Recognition error	FR-IS-04	Blurred or unrecognised image, retry and alternative-mode exits, confidence threshold (open item)	Negative · Exception
SS-SCR-017	Voice search — Listening	FR-VS-01	Microphone permission, waveform feedback, auto-stop threshold, manual cancel	Functional · Negative
SS-SCR-018	Voice search — Results	FR-VS-02	Transcribed keyword echoed back and editable, result set identical to typing the same keyword	Functional · Integration
SS-SCR-019	Voice search — Speech error	FR-VS-03	Unclear speech, silence, background noise; retry and fall back to typing	Negative · Exception

2.8.3 Rule-based test conditions

The storyboard states explicit thresholds and data rules. Each one below is a pass / fail assertion and must have at least one test case.

No	Rule / threshold	Assertion	Screen
1	Autocomplete debounce	No suggestion API call is issued until 300 ms after the last keystroke	SS-SCR-002
2	Loading display threshold	Processing screen appears only when the response exceeds 300 ms	SS-SCR-004
3	Timeout threshold	Beyond 5 s the screen changes to a network-error message with a Retry action; a blank screen is a defect	SS-SCR-004
4	Recent keywords	Maximum 10 chips, most recent first; the whole block is hidden when history is empty; "Clear all" removes every entry	SS-SCR-001
5	Trending keywords	Top 10 from the last 7 days, normalised, profanity-filtered, and only keywords that return more than zero results	SS-SCR-001
6	Frequently purchased	Logged-in only. Source: order history with status COMPLETED in the last 90 days; Top 5–10 SKUs with stock > 0	SS-SCR-001
7	No auto-replace	A misspelled keyword is never silently replaced; the typed string stays in the input and the correction is offered as a link	SS-SCR-003
8	Sponsored transparency	The Sponsored label renders in every view — grid, list, cross-sell, image results and voice results	SS-SCR-005 ... 007 · 015 · 018
9	Zero-result prevention	No search path ends in an empty page; at least one of the three recovery tiers is always shown	SS-SCR-003 · 012 · 016 · 019
10	Mode parity	A keyword reached by voice or image produces the same result set, sort, filter and add-to-cart behaviour as typing it	SS-SCR-015 · 018

2.8.4 Out of scope for Smart Search

- Product detail page (PDP), Cart and Checkout screens — reached from search results but tested under their own scope.
- Login and signup — covered by section 2.9.
- Back-office keyword configuration and Retail Media / sponsored-slot administration.
- Technical specification of the search engine itself (indexing, embedding, ranking implementation) — a separate document.
- The Web / mWeb layout of Smart Search. The storyboard covers the Mobile App only; Web inherits the same logic and is added to this plan when its storyboard is issued.

2.9 Simplified User Flow — Requirement-Level Test Scope (SUF detail)

Section 2.6 states the SUF scope as 7 features in 2 feature groups, taken from the Feature Baseline. MART-AN02-SRS-Simplify User Flow-v0.1-20260814 has since specified the module end to end — from first app launch to a placed order — as 6 sequential business areas containing 54 functional requirements, 12 business requirements, 34 business rules, 10 process flows, 26 screens and 5 non-functional requirements. This section restates the SUF test scope at that level. Where section 2.6 and this section differ, this section governs.

2.9.1 Business areas in scope

Code	Business area	FR	ID range	Screens	Test focus
ONB	Onboarding	7	ONB-01 ... 07	4	Language, three system permissions, mandatory store selection, introduction carousel
HP	Homepage	10	HP-01 ... 10	6 (+1 gate)	Header context, address create / change, service-zone and ETA resolution, the two login gates, store change, logout
SU	Signup	9	SU-01 ... 09	3	Phone, OTP channel (Zalo / SMS), OTP entry, PIN creation, profile, account creation, biometric offer, resume of the original action
LI	Login	7	LI-01 ... 07	2	Priority-ordered authentication (biometric → PIN → OTP), resume, PIN reset, attempt limit, device change
CT	Cart	11	CT-01 ... 11	7	Delivery / Pickup selection, service and slot rules, freeship top-up, product management, promotions, coupons, store change, lockers, checkout hand-off
CO	Checkout	10	CO-01 ... 10	3	Locked order review, note and packing, gift card, coupons and points, payment method, VAT invoice, policies, place order
Total	6 areas	54	—	26	10 process flows · 12 business requirements · 34 business rules · 5 NFR

2.9.2 Process-flow test scenarios

Each process flow is one end-to-end test scenario. Every branch inside a flow is a separate test case.

Flow	Scenario	Trigger / pre-condition	Requirements realised
PF-01	Master end-to-end journey	First app launch → order placed	BR-01 ... BR-04
PF-02	Onboarding	First launch, no store selected	ONB-01 ... ONB-07
PF-03	Homepage context resolution	Guest on Homepage, address entered	HP-01 ... HP-05, HP-08 ... HP-10
PF-04	Login gate	Add to Cart, or Cart icon tapped, as Guest	HP-06, HP-07
PF-05	Signup	Phone number not yet registered	SU-01 ... SU-09
PF-06	Login	Registered phone number	LI-01 ... LI-07
PF-07	Cart — Delivery	Member with items in cart, address in a served zone	CT-01 ... CT-08, CT-11
PF-08	Cart — Pickup	Member selects Pickup at a store	CT-02, CT-09, CT-10
PF-09	Store change inside the Cart	Serving store changed while items are in the cart	CT-07
PF-10	Checkout	Cart handed off to Checkout	CO-01 ... CO-10

2.9.3 Branch and negative test conditions

Ref	Condition	Expected system behaviour	Source
B1	Location permission denied at Onboarding	No default store is pre-selected; the customer selects a store manually; the flow does not stop	ONB-05 · BRL-02
B2	Address outside every service area	No ETA and no delivery date shown; "This area is not served yet" plus a Change address CTA; browsing and Pickup continue normally	HP-04 · BRL-11
B3	Customer cancels at the	Return to Homepage as Guest; the product is not	HP-06 · HP-07

	login gate	added and the Cart does not open	
B4	Phone number already registered on Signup	Error shown, the number is retained, and the CTA switches to Login	SU-09
B5	Serving store changed while in the Cart	One consolidated warning listing every impact; the customer confirms Continue or Cancel	CT-07 · BRL-24
B6	Session expired at a login gate	Handled as a Guest — assumption carried in HP-07, pending OQ-04	HP-07
B7	Wrong OTP or PIN entered repeatedly	Attempt limit and lock behaviour enforced; PIN lock forces OTP re-verification	NFR-01 · LI-06
B8	Item goes out of stock while in the cart	Out-of-stock items are separated from the buyable group and cannot be checked out	CT-05 · CT-07
B9	Customer edits the order at Checkout	The order is read-only; any change returns the customer to the Cart with selections preserved	CO-01 · BRL-31 · NFR-04
B10	Guest address created before first login	Address is stored on the device only and is synced to the account at first login	HP-05

2.9.4 Business-rule assertion groups

Group	Rules	What must be asserted
Access & authentication	BRL-05 ... BRL-08 · BRL-13 ... BRL-17	Login is enforced at exactly two points and nowhere else; the original action is always resumed after authentication; OTP is sent over Zalo first and never falls back automatically
Store, address & service zone	BRL-01 ... BRL-04 · BRL-09 ... BRL-12	Store selection drives catalogue, price, stock and promotion; service zone is resolved from the address; an unserved address shows no estimate at all
Delivery, slot, bulky & fee	BRL-18 ... BRL-23 · BRL-28 ... BRL-30	Freeship thresholds, 60-Minute / Standard / Next-day eligibility, bulky-order rule (> 15 kg or a bulky-labelled item), earliest selectable day T+2 / T+4, slot capacity
Cart & promotion	BRL-24 · BRL-32 · BRL-34	Consolidated store-change warning; delivery coupons and order coupons grouped separately and applied automatically; out-of-stock gift-item handling
Checkout integrity	BRL-31 · BRL-33	The order total, promotion set and price cannot change between review and payment

2.9.5 Screen inventory

"Redesign" means the screen exists today on lottemart.vn and its behaviour changes — regression against the current site is required. "New" means it does not exist in the As-Is flow.

Screen ID	Screen name	Area	Realises	Type
SCR-ONB-01	Onboarding — Language selection	ONB	ONB-01	New
SCR-ONB-02	Onboarding — System permission requests	ONB	ONB-02 ... 04	New
SCR-ONB-03	Onboarding — Store selection (mandatory)	ONB	ONB-05	New
SCR-ONB-04	Onboarding — Introduction carousel	ONB	ONB-06	New
SCR-HOM-01	Homepage — Guest, store context	HP	HP-01, HP-04, HP-10	Redesign
SCR-HOM-02	Homepage — Create delivery address	HP	HP-02, HP-03, HP-05	Redesign
SCR-HOM-03	Homepage — Change address	HP	HP-09	New
SCR-HOM-04	Homepage — Change store	HP	HP-08	Redesign
SCR-HOM-05	Homepage — Delivery estimate states	HP	HP-04	New
SCR-HOM-06	Homepage — Member state & logout	HP	HP-01, HP-10	Redesign
SCR-AUT-01	Login gate — Register or sign in	HP	HP-06, HP-07	New
SCR-AUT-02	Signup — Phone number & OTP	SU	SU-01 ... 03, SU-09	Redesign
SCR-AUT-03	Signup — Create PIN	SU	SU-04	New
SCR-AUT-04	Signup — Profile & biometric offer	SU	SU-05 ... 08	Redesign
SCR-AUT-05	Login — Authenticate	LI	LI-01 ... 04, LI-06	Redesign

SCR-AUT-06	Login — Forgot PIN & reset	LI	LI-05	New
SCR-CRT-01	Cart — Standard delivery	CT	CT-01 ... 06	Redesign
SCR-CRT-02	Cart — 60-Minute delivery	CT	CT-08	New
SCR-CRT-03	Cart — Next-day delivery	CT	CT-08	New
SCR-CRT-04	Cart — Pickup & locker	CT	CT-02, CT-09, CT-10	Redesign
SCR-CRT-05	Cart — Item limit & remove confirmation	CT	CT-05	New
SCR-CRT-06	Cart — Store change warning & out of stock	CT	CT-07	New
SCR-CRT-07	Cart — Slot and store selection sheets	CT	CT-08	Redesign
SCR-CHK-01	Checkout — Locked order review	CO	CO-01 ... 03	Redesign
SCR-CHK-02	Checkout — Payment, points & summary	CO	CO-04 ... 10	Redesign
SCR-CHK-03	Checkout — Pickup variant & compact list	CO	CO-01	New

2.9.6 Non-functional requirements in scope

ID	Category	Requirement to be tested	Status in the SRS
NFR-01	Security	Wrong-attempt limit on both OTP and PIN: temporary lock after N wrong OTP entries, a cap on OTP requests per period, PIN lock after N failures forcing OTP re-verification	Assumption — value pending (OQ-03)
NFR-02	Security	One device holds one active biometric ticket at a time; overwriting it between accounts requires an explicit warning	Proposed — OQ-07, OQ-11
NFR-03	Privacy	The delivery address is cleared on logout; the selected store is retained	Agreed
NFR-04	Data integrity	The order is locked read-only at Checkout so price, promotion set and total cannot drift between review and payment	Agreed
NFR-05	Cost control	OTP is sent over Zalo before SMS and never falls back to SMS automatically	Agreed

NFR-01 has no agreed threshold at the time of writing. Its test cases can be authored but cannot be given pass / fail criteria until OQ-03 is closed — tracked in section 13.

2.9.7 Out of scope for Simplified User Flow

- Post-order fulfilment — picking, packing and dispatch (OMS / DMS workstream).
- Order tracking and delivery status (Order Management module).
- Customer service, returns and refunds.
- Product listing, product detail and category browsing — covered by the Homepage and Product Card deliverables (DE02 / DE03).
- Search behaviour — covered by section 2.8.
- Recommendation engine behaviour — covered by DE05 Recommendation Strategy.
- Loyalty tier and point accrual calculation — consumed here, defined in the membership workstream.
- Back-office configuration of store radius, slot capacity and thresholds — consumed as configuration, not tested as screens.

2.10 Out of Scope

- RECIPE — AI Recipe Shopping. Removed from the Part 1 scope; it is not covered by this Test Plan and has no test cases, traceability rows or schedule allocation here.
- Any module or feature not listed in sections 2.1 to 2.9.
- Module-specific exclusions are stated in sections 2.8.4 (Smart Search) and 2.9.7 (Simplified User Flow); they are not repeated here.
- Non-functional targets that have no agreed acceptance threshold at the time of writing — see section 13, Open Items.

3. Assumptions

- Backend AI and ML services required by AI Search, AI Recommendation and Personalized Homepage are available and stable in the test environment.
- Product catalog, pricing, promotion and customer profile master data meet a minimum completeness standard before test execution begins, across all six modules.
- Test users, accounts and any required customer-segment, subscription or purchase-history seed data are pre-provisioned before test execution.
- Where modules share underlying logic — for example the personalization and ranking logic used by both AI Search and Personalized Homepage — that logic is a single shared service rather than duplicated per module.
- The Smart Search storyboard and the Simplified User Flow SRS are both at v0.1 and are not yet baselined. Test cases authored against them are provisional until each document is approved; any change is handled as a test-design change, not a defect.
- The Mobile App build under test implements the same logic as the Web and mWeb builds. Where the Smart Search storyboard specifies the Mobile App only, Web coverage is added once the Web storyboard is issued.
- Environment, infrastructure and test tooling are TBD at this stage and are to be confirmed with the Project Manager before the Testing phase begins.

4. Dependencies

- MART Online Renewal 2026 Scope and Tasks is the source of truth for in-scope features across all six modules, until a formal BRD is issued per module.
- Product catalog and OMS data feed for product, price, inventory and promotion information.
- Customer profile, purchase-history and subscription data source for personalization, recommendation and subscription-management logic.
- MART-DE03-Smart Search Storyboard-Mobile App-v0.1-20260816 — source of the screen-level Smart Search scope in section 2.8 and the traceability rows in 11.2.
- MART-AN02-SRS-Simplify User Flow-v0.1-20260814 — source of the requirement-level Simplified User Flow scope in section 2.9 and the traceability rows in 11.3.
- Test environment with all six modules deployed and reachable (environment TBD).

5. Test Environment

The environment and tooling below are not yet confirmed. Each TBD is tracked in section 13, Open Items.

Component	Configuration / detail
Application	All six Part 1 modules — TBD (Web / mWeb / App)
Environment	TBD — to be confirmed with PM and Tech Lead
Test data	Catalog, customer profile, purchase history and subscription seed data — provisioned before execution
Test tools	JMeter (performance), Postman (API and integration)
Defect tracking	TBD — to be confirmed with PM

6. Test Approach

Testing is organised by the test types below. Each authored test case is tagged with a coverage category so that traceability and completeness can be measured per requirement.

Test type	Approach
Functional	Execute positive end-to-end flows against the acceptance criteria across all six modules: AI Search, AI Recommendation, Subscription Service, Personalized Homepage, Simplified User Flow and Social Commerce.

Validation	Verify mandatory-field, format and business-rule validation on all input forms and requests, in every module.
Negative	Attempt disallowed or invalid actions and confirm correct rejection and handling, in every module.
Exception / Error	Simulate backend service unavailability and timeouts, and confirm clear error handling, per module.
Security / RBAC	Verify access control is enforced where a module exposes role- or segment-specific behaviour, for example subscription tier or customer segment.
Integration	Validate data contracts with the catalog / OMS and customer-profile services each module depends on, including cross-module consistency where logic is shared.
Performance	Measure response time and load behaviour for each module under concurrent users (targets TBD).
Multimodal input	Exercise text, image and voice entry paths for the same shopping intent and confirm they converge on the same result set, sort, filter and add-to-cart behaviour (sections 2.8.1, 2.8.3).
Flow / journey	Execute each process flow PF-01 to PF-10 end to end and each documented branch B1 to B10 as a separate case (section 2.9.2, 2.9.3).
Regression	For every screen marked "Redesign" in section 2.9.5, compare the To-Be behaviour against the current lottemart.vn behaviour and confirm the change is the intended one.

7. Entry & Exit Criteria

7.1 Entry Criteria

- All six modules deployed and reachable in the test environment (environment TBD).
- Test users, roles and any required seed data — purchase history, subscription state, catalog entries — provisioned for every module.
- Requirements baseline reviewed and approved: 35 feature groups across six modules, plus the 19 Smart Search screens (section 2.8) and the 54 Simplified User Flow requirements over 26 screens (section 2.9). Test cases traced to requirements.
- Camera, photo-library, microphone and location permissions grantable on every test device, and OTP delivery over Zalo and SMS available in the test environment.
- Backend and AI services each module depends on are available, or stubbed with agreed contracts.

7.2 Exit Criteria

- 100% of planned test cases executed across all six modules.
- No open Critical or High severity defects in any module; Medium and Low defects triaged with agreed dispositions.
- All High-priority requirements, in every module, have at least one passing traced test case.
- No search path ends in a blank screen, and no Simplified User Flow branch (B1 to B10) is left unexecuted.
- Cross-module consistency checks pass where personalization or ranking logic is shared, for example AI Search versus Personalized Homepage.

8. Test Schedule

All start and end dates below are TBD pending confirmation with the Project Manager. Modules may be scheduled in parallel or in sequence.

Phase	Activity	Start	End	Owner
Preparation	Requirement review, feature-baseline walkthrough across all six modules, test-data seeding, environment smoke test	TBD	TBD	BA, PM
Test Design	Author and review test cases and the traceability matrix per module and feature	TBD	TBD	QC Team

Functional Execution	Execute functional, validation and negative cases across all in-scope screens, per module	TBD	TBD	QC Team
Integration & Exception	Execute integration cases and exception handling for dependent services, per module	TBD	TBD	QC Team
Performance	Execute performance cases, based on the agreed requirement targets	TBD	TBD	QC Team
Closure	Defect retest, result consolidation, exit review and sign-off across all six modules	TBD	TBD	QC Team / PM

9. Risks & Mitigation

ID	Risk	Impact	Likelihood	Mitigation
R-01	Backend and AI services (search, recommendation, personalization) unstable or unavailable during the test window	High	Medium	Agree service availability windows up front; use stubs with contract parity where possible
R-02	Timeline, environment and test tooling not yet confirmed, risking schedule slippage across all six modules	Medium	High	Confirm timeline, environment and tooling with the PM before the Testing phase begins
R-03	Catalog, customer and subscription data quality (completeness, tagging) affects test result validity	Medium	Medium	Confirm the minimum data-completeness standard with the BA before the Testing phase begins
R-04	Overlapping AI and personalization logic across modules (AI Search, AI Recommendation, Personalized Homepage) diverges in ranking behaviour	Medium	Medium	Confirm with BA and PM whether ranking and personalization is one shared service; add explicit cross-module consistency test cases if so

Any risk that materialises is transferred to PM05 Issue & Risk Register and managed there.

10. Test Deliverables

- MART-TE01-Test Plan-Part 1-v0.3-20260817.docx — this consolidated Test Plan.
- Test case set and requirement traceability matrix, per module.
- TE02 Integration Test Result — execution results, pass rate and retest outcome.
- Defect log with severity, owner, status and closure record.
- Closure and sign-off report against the exit criteria in section 7.2.

11. Requirements & Traceability

Traceability is kept at three levels. 11.1 holds the 35 feature-level requirements that cover all six Part 1 modules. 11.2 and 11.3 add the screen and requirement level for the two modules that now have a specification. The test-case count is completed during Test Design and kept current until closure; every High-priority requirement must have at least one passing traced test case before the exit criteria in section 7.2 can be met.

11.1 Part 1 modules — feature-level requirements

Req ID	Requirement	Priority	Source	Test cases	Coverage status
REQ-AISCH-01	Search Intent	High	AISCH — Feature Baseline		
REQ-AISCH-02	Visual Search	High	AISCH — Feature Baseline		
REQ-AISCH-03	Multi-language Search	Medium	AISCH — Feature Baseline		

REQ-AISCH-04	Search Result	High	AISCH — Feature Baseline		
REQ-AISCH-05	Search Result Sponsor Display	High	AISCH — Feature Baseline		
REQ-AISCH-06	Product Recommendation	Medium	AISCH — Feature Baseline		
REQ-AIREC-01	AI Product Review Summary	High	AIREC — Feature Baseline		
REQ-AIREC-02	Notification Suggestion	Medium	AIREC — Feature Baseline		
REQ-AIREC-03	Cart Cross-sell	Medium	AIREC — Feature Baseline		
REQ-AIREC-04	Coupons Suggestion	Medium	AIREC — Feature Baseline		
REQ-AIREC-05	Product Suggestion	High	AIREC — Feature Baseline		
REQ-AIREC-06	Shopping Suggestion	High	AIREC — Feature Baseline		
REQ-SUBS-01	Subscription Signup & Payment	High	SUBS — Feature Baseline		
REQ-SUBS-02	Subscription Activation	High	SUBS — Feature Baseline		
REQ-SUBS-03	Subscription Management	High	SUBS — Feature Baseline		
REQ-SUBS-04	Subscription Auto-Renewal	High	SUBS — Feature Baseline		
REQ-SUBS-05	Subscription Benefits	High	SUBS — Feature Baseline		
REQ-SUBS-06	Subscriber List Management	High	SUBS — Feature Baseline		
REQ-SUBS-07	Subscription Refund Management	High	SUBS — Feature Baseline		
REQ-SUBS-08	Subscription Notification	High	SUBS — Feature Baseline		
REQ-HOME-01	Hero Welcome + Search	High	HOME — Feature Baseline		
REQ-HOME-02	Personalized Recommendation Feed	High	HOME — Feature Baseline		
REQ-HOME-03	Deals / Price Shock Section	High	HOME — Feature Baseline		
REQ-HOME-04	Membership CTA Block	High	HOME — Feature Baseline		
REQ-HOME-05	Content (Live / Video)	High	HOME — Feature Baseline		
REQ-HOME-06	Meal / Lifestyle Inspiration	High	HOME — Feature Baseline		
REQ-HOME-07	Personalized Banner Recommendation	High	HOME — Feature Baseline		
REQ-HOME-08	Personalized Product Recommendation	Medium	HOME — Feature Baseline		
REQ-HOME-09	Homepage Structure & Zone Management	Medium	HOME — Feature Baseline		
REQ-HOME-10	Content Scheduling & Priority	High	HOME — Feature Baseline		
REQ-SUF-01	Simplified Purchase Process	High	SUF — Feature Baseline		
REQ-SUF-02	Simplified Signup	High	SUF — Feature		

			Baseline		
REQ-SOCIAL-01	Short-Video Feed	High	SOCIAL — Feature Baseline		
REQ-SOCIAL-02	Short-Video Management	High	SOCIAL — Feature Baseline		
REQ-SOCIAL-03	Short-Video Analytics	High	SOCIAL — Feature Baseline		

11.2 Smart Search — screen-level coverage (source: DE03 Storyboard v0.1)

Req ID	Source REQ ID	Requirement	Priority	Screens
REQ-SS-01	FR-SP-02	Pre-search entry: recent, trending, frequently purchased	High	SS-SCR-001
REQ-SS-02	FR-SP-01	Autocomplete suggestions with 300 ms debounce	High	SS-SCR-002
REQ-SS-03	FR-SP-03	Spell correction, suggest-only (no auto-replace)	High	SS-SCR-003
REQ-SS-04	NFR-SS-01	Processing state and timeout handling (shared by 3 modes)	High	SS-SCR-004
REQ-SS-05	FR-PR-01 / 02	Result ranking and grid / list presentation	High	SS-SCR-005, 006
REQ-SS-06	FR-PR-05	Cross-sell block on the results page	Medium	SS-SCR-007
REQ-SS-07	FR-SF-01	Sort	High	SS-SCR-008
REQ-SS-08	FR-SF-02 / 03	Filter sheet and post-filter result state	High	SS-SCR-009, 010
REQ-SS-09	FR-AC-01	Add to cart from any result surface	High	SS-SCR-011
REQ-SS-10	FR-ZR-02	Zero-result three-tier recovery	High	SS-SCR-012
REQ-SS-11	FR-IS-01 ... 04	Image search: capture, gallery, results, error	High	SS-SCR-013 ... 016
REQ-SS-12	FR-VS-01 ... 03	Voice search: listening, results, error	High	SS-SCR-017 ... 019

REQ IDs FR-SP-01 / 02, FR-SF-01, FR-PR-01 / 02 / 05 and FR-ZR-02 come from the Figma Logic section; the remainder were coined by the storyboard and must be reconciled with the Smart Search BRD before this traceability is baselined — Open Item 9.

11.3 Simplified User Flow — requirement-level coverage (source: AN02 SRS v0.1)

Req ID	Source FR	Requirement	Priority	Screens	Flow
REQ-SUF-01	ONB-01 ... 07	Onboarding: language, permissions, mandatory store, carousel	Must / Should	SCR-ONB-01 ... 04	PF-02
REQ-SUF-02	HP-01 ... 05	Homepage context: header, address lifecycle, delivery estimate	Must	SCR-HOM-01, 02, 05	PF-03
REQ-SUF-03	HP-06, HP-07	Login gate at Add to Cart and at Cart entry	Must	SCR-AUT-01	PF-04
REQ-SUF-04	HP-08 ... 10	Store change, address change, logout	Must	SCR-HOM-03, 04, 06	PF-03
REQ-SUF-05	SU-01 ... 09	Signup: phone, OTP channel, OTP, PIN, profile, biometric, resume	Must	SCR-AUT-02 ... 04	PF-05
REQ-SUF-06	LI-01 ... 07	Login: biometric → PIN → OTP, reset, attempt limit, device change	Must	SCR-AUT-05, 06	PF-06
REQ-SUF-07	CT-01 ... 06	Cart: entry, Delivery / Pickup, freeship top-up, filters, products, promotions	Must	SCR-CRT-01, 05	PF-07
REQ-SUF-08	CT-07	Store change inside the Cart — consolidated warning	Must	SCR-CRT-06	PF-09
REQ-SUF-09	CT-08	Service, slot and Next-day rules	Must	SCR-CRT-02, 03, 07	PF-07
REQ-SUF-10	CT-09, CT-10	Pickup rules and lockers	Should	SCR-CRT-04	PF-08

REQ-SUF-11	CT-11	Checkout hand-off	Must	SCR-CRT-01	PF-07
REQ-SUF-12	CO-01 ... 03	Locked order review, note, packing	Must	SCR-CHK-01, 03	PF-10
REQ-SUF-13	CO-04 ... 07	Gift card, coupons and points, payment method, VAT invoice	Must	SCR-CHK-02	PF-10
REQ-SUF-14	CO-08 ... 10	Final summary, policies, place order	Must	SCR-CHK-02	PF-10
REQ-SUF-15	NFR-01 ... 05	Security, privacy, data integrity and cost-control requirements	Must	SCR-AUT-02, 05 · SCR-CHK-01	PF-05, 06, 10

Must / Should priorities are a BA proposal in the SRS and are pending Product Owner confirmation (OQ-24) — Open Item 17.

12. Roles & Responsibilities

Role	Responsibility
Business Analyst	Owns the requirements; reviews test coverage and approves traceability.
QC Team	Owns this test plan, the schedule and the entry / exit gates. Authors and executes test cases, raises defects and manages the defect life cycle.
Developer	Fixes defects, and supports the environment and interface stubs.
Project Manager	Approves go / no-go against the exit criteria and manages stakeholder sign-off.

13. Open Items

Every item below is unresolved at v0.2. Each one blocks either the schedule or the acceptance of a result, and must be closed before the Testing phase begins.

No	Open item	Owner	Decision needed by	Impact if unresolved
1	Test schedule — start and end dates for all six phases in section 8	PM	Before Test Design starts	Test Design and resourcing cannot be committed
2	Test environment — URL, deployment target and access for the six modules	PM / Tech Lead	Before Preparation starts	No environment smoke test, execution cannot start
3	Defect tracking tool and workflow	PM / QC	Before execution starts	Defects cannot be raised or tracked consistently
4	Performance targets — response time and concurrent-user thresholds per module	BA / Tech Lead	Before Test Design starts	Performance cases have no pass / fail criterion
5	Minimum data-completeness standard for catalog, customer and subscription data	BA	Before Preparation starts	Personalization and recommendation results cannot be judged valid
6	Whether ranking and personalization is one shared service across modules	BA / Tech Lead	Before Test Design starts	Cross-module consistency cases cannot be scoped (risk R-04)
7	BA owner for the Part 1 test effort	PM	Before Test Design starts	No named reviewer for traceability approval
8	Acceptance criteria for AI output quality — relevance and ranking accuracy	BA / QC	Before Test Design starts	AI features can only be smoke-tested, not accepted
9	Smart Search REQ-ID reconciliation — several FR IDs were coined by the storyboard and do not exist in a BRD	BA	Before Test Design starts	Traceability in 11.2 cannot be baselined; duplicate or orphan requirements
10	Technical thresholds for Smart Search — fuzzy-match distance, image-recognition confidence, voice auto-stop	BA / Tech Lead	Before Test Design starts	Cases for SS-SCR-003, 016 and 019 have no pass / fail criteria

11	Are active filters retained when the customer enters a new keyword	BA / Design	Before Test Design starts	Expected result for SS-SCR-010 is undefined
12	"Frequently purchased" behaviour for guests — hidden entirely, or replaced by another block	LOTTE MART / PO	Before execution starts	Guest path on SS-SCR-001 cannot be verified
13	Sponsored slot placement rules — fixed position or ranking-dependent	LOTTE MART / BA	Before Test Design starts	Ad-transparency assertion in 2.8.3 cannot be made exact
14	Two Zero-Result blocks are hidden in the current design ("similar keywords", "you may need")	Design	Before Test Design starts	Tier 2 of the zero-result recovery cannot be tested
15	OTP and PIN wrong-attempt limits and lock duration (SRS OQ-03)	Security / Dev	Before Test Design starts	NFR-01 has no threshold; B7 has no pass / fail criteria
16	Which system owns slot capacity and Next-day date configuration (SRS OQ-12)	SI / Dev	Before Preparation starts	CT-08 integration cases cannot be scoped
17	Confirmation of Must / Should priorities across the Simplified User Flow (SRS OQ-24)	PO	Before Test Design starts	Priority column in 11.3 is provisional; exit criteria on High-priority coverage cannot be applied
18	Complete locker rules — selection, holding period, cost, no-show (SRS OQ-23)	LOTTE MART	Before Test Design starts	SCR-CRT-04 state B cannot be tested
19	Carton packing fee and the complete promotion-type catalogue (SRS OQ-16, OQ-17)	LOTTE MART	Before Test Design starts	CO-08 total calculation and CT-06 promotion cases cannot be verified
20	Web / mWeb storyboard for Smart Search	BA / Design	Before Web execution starts	Only the Mobile App is covered by section 2.8; Web has no screen baseline